

Time travel

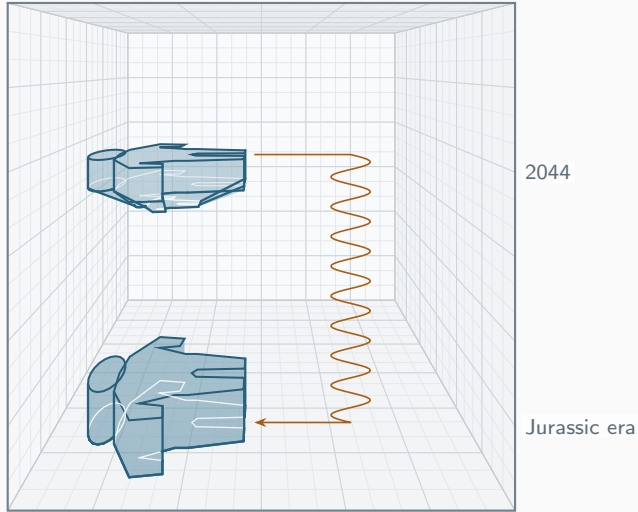
PHIL UA-78 Metaphysics

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Lewis's defence of the possibility of time travel

A time travel story



Does a single person live at both times?

According to Lewis, a person is an aggregate of person-stages which can be ordered in such a way that the stages that come later in the order have psychological properties that are similar enough to, and caused in the right way by, the psychological properties of the stages immediately prior to them in this order (and in which no two stages are too dissimilar psychologically).

On this view, the stages in our story can compose a single person, provided we have an appropriate pattern of *backwards causation*, whereby events in 2044 cause events in the Jurassic.

Backwards causation

Given this account, the key question is whether we can have *backwards causation* in such cases.

Indeed, stories of backwards causation without time travel (e.g. people who can see the future) introduce the same temptation towards inconsistency (or restorations of consistency that invoke multiple 'time streams') as stories of backward time travel.

Let's consider what counterfactual theories of causation like Lewis's say about the possibility of backwards causation.

Backwards causation and backwards counterfactual dependence

For theories like Lewis's, counterfactual dependence is sufficient for causation (so long as the counterfactual is read in the 'standard' as opposed to 'backtracking' way).

So there will be such backwards causation provided that the closest worlds where things are different in relevant ways in 2044 are worlds where they are different in corresponding ways in the Jurassic.

So long as we can have laws of nature according to which in some cases the state at some earlier time has to match the state at some later time, the theory of closeness from Counterfactual Dependence and Time's Arrow allows for this. (Nevertheless, this theory might need some modification to get intuitive results about how counterfactuals work in the relevant worlds!)

Rejecting backwards counterfactual dependence

You could try to get off the boat here by adopting an account of counterfactuals on which backwards counterfactual dependence of the relevant kind is impossible.

But if there can be laws according to which the conditions at either end of a time gate are required to match, this is hard to square with intuitive ideas about the importance of laws in the evaluation of counterfactuals!

So, you'll probably also have to say that laws that work like that are impossible.